mages too. - **Fidelity**: `isFaithful(sheet)` e
t as works stay readable for machines. T
. The filler contains words like *official*, wh
all `<text>` content, whitespace normalised).
ity | | Image only | `tesseract` on a PNG render
—|—|—|—|—| | high | clear | exact | exact |
ge model (first sample) Classic OCR knows only
the word order cannot be known in advance. Me
um | 65 | 100 % | | fras | 194 | 100 % | This is small sa
Fe.** Chrome set "ffi" in *official* as one ligature gly
, and `pdftotext`'s layout mode moved the very first w
Medium** Filter the ornamental **Memphis-mokey**
draw -F txt`) returns every visible character. The cause
e the PDF text layer. The SVG master keeps them; the PD
cters read in the one-per-complete Measure the ratio or
ng order | | pdfminer.six | 70 of 70 | | | pypdf | 10 of 10 | reads a
ers above that lose most of their text (10–30 % read). Its defau
ce), this holds even for the EU AI Act at 0.5 mm type. White-s
e as not exact. The agent library avoids arrows for that reason.
stored order; with legibility *high* it is exact for layout-reconstr
than 60 000 small characters. Tools that should the visual hyper
reproducible numbers are very welcome; open an issue or pull requ
gs change the picture. All numbe come from `scripts/size-study.py`
chts • width ± type size** (mains used as a reference). White-s
lines • 7.2 B × runs • 0.9 B × letter
ved without visible change | Step
one empty `/MonStroke1` element
set to (title or motif card), both b
change to full | |—|—|—|—|—| |
S, one width | 1.93 MB | 40 | 0.942 | |
% | | colour step 96, one width | 2.6
s brightness / colour | change to f
| | 4 weights, one width | 0.58 MB | |
5 | 0.30 % | | colour step 96, one wi
rtal streaks; with one size the photo motif follows its picture more closely (0.965 inste
- **Width variants are not worth it.** Dropping condensed and expanded changes so
e.mjs`); `--weights 2` makes them smaller still. In the app, the font-cuts slider limits th
evestudy.mjs --evaluate` measures the PDFs of the last run again. `--- <!-- docs/hidde
e key. ## How it works It is a book cipher over the work's public text. 1. **The public t
ith MuPDF). 2. **Hiding:** for every character of the secret, all its occurrences in the publ
and refused instead of predicting garbage. 6. **Browser:** the app runs entirely in the browser
ct on top of a short header: measured on a 47 000-character text, 26 characters of secret ga
ecret, so it is better not to put it on the command line, where it shows up in the process list an
`org/app), the panel **Hidden message** does the same for the current work's text. `hideSecr
e secret.** Whoever has the key and the work can read the message. Share the key over anothe
e the message must stay secret even from someone who may get the key: then encrypt the messa
based from short ones. - The key fits exactly one text. Re-setting the same text into a new picture keep
its source and the key made from t docs/ai-readability.md`). ## Hone
md --> ## Passwords, keys and seed d. Everything is open source; ev
s/key-management.md) and [cipher-
e did not invent a cipher, and we look
efuse to publish a record that is wrong
and nobody can help you. - **It is not
typographics-art.org/docs/struc
passed by file, standard input or environm
work binding (`tgx2.`), key splitting (XOR, Shamir) a
(https://typographics-art.org/docs/cipher-review.md) (w
ccs/passwords-and-keys.de.md)](https://typographics-art.or
nown", not "why only can ever get at it". - a lost key loses the data, a c
sealed so it only opens with the key, and an **adapter** (`cli/tg-adap
anything being uploaded. Large media is a seekable `tgst1` stream file (the
phics-art.org/docs/encrypted-media-adapter.md). An internal adversarial r
curity-review-internal-2026-10-b.md](https://typographics-art.org/docs/secur
text. ** The work is published as real text: HTML, SVG, or PDF with a text layer. Ext
orks ``` mermaid flowchart LR T[Text] --> M[Measure every glyph-cho every wid
advance width and ink coverage at ten weights (and, if the optional width axis is on,
ed on to the next glyph and the next line (error diffusion), so tones average out correctly
org/docs/algorithms.md)](https://typographics-art.org/docs/algorithms.md). ## Use it ##
r browser and never uploaded; the s
ork.html) shows a finished work in thr
) CLI: Node ≥ 22; API: Node ≥ 22.5. `
/ char | word (one style per word: calm
ts URL (image only) ``` The pure mod
OME_PATH` If it is not found): PDFs also
``` The output format follows the fil
lf-hosting: [api/README.md](https://gi
dgar Allan Poe | | <br>**Fail-not-Yanus** :
ontiguous passage of a public-domain w
mb). Each preview links to
e of the sitter's biography, a city made o
r VoiceOver. - **Findable art.** Searc
s) arranges text along a shape, usually by
y variable fonts?** A weight axis (and s
he texts and photos you use. ## Contr
hub.com/typ0graphic5/tg-art/blob/main
ONTRIBUTING.md); please follow
m) # only to deploy the website: t
-|| `src/` | the library | | `cli/` | comman
ements and rules | | `about/` | gener
, the project's own texts and motifs an
ng a picture from a given text by settin
t follows three rules. ### 1. Fidelity
gap at a space (or indent a line) so that
en (a URL, a base64 string, the dot-join
an unsharp mask (**emphasise edges**
× (1 − grey) × lineRatio / 0.92 ``` For eve
ext to full-size ones read like stray cap
dge tests the tone of the side by side a
heet (3 000 characters make about 42 l
scanned in steps of 0.3 font sizes; stretches wh
Hearn • Hokusai | <br>**The Starr • H. G. Wells • Hub
pt the Universal Declaration of Human Rights
, which is the work itself: op
y. Any reader, human or machine, gets both
and AI crawlers index the words, not just a
pographics Art keeps a normal reading laya
a width axis) gives each glyph a continuou
pographics Art is meant to become a share
F_CONDUCT.md). ``` sh npm test # node --t
he site node demo/moby-dick/build.mjs z
ation builders | | `gallery/` | the 15 works | | `e
or [docs/self-hosting.md](https://typographic
heir own licences: see [NOTICE.md](https://
es of each character. It is the count
placed, inserted, recorded or drop
not even a space, is inserted, and t
** (for example Lorem ipsum to rea
e the source text. It applies not
ets fragments; one that reads by line
n the set form, so that extractors see it once. Not yet listened to with NVDA, JAWS or Voic
and word spacing), including wide gaps at spaces and indents that leave parts of the
ad by zooming in or by machines. Implementations should offer a legibility setting th
nvisible; the message lives only in the key. Tiny or Invisible text must never be used
The pipeline lives in `src/`. Everything except `browser.js` runs in Node
the share of the glyph's cell (advance × line height) covered by ink, measured by
would silently measure a fallback font. ## 2. Tone map (`tone.js`) The ima
mand and mean-color of any rectangle an O(1) lookup. ## 3. Glyph selection (rule
us glyph) to the cheapest candidate; ``` cost = (eff − target)² + 0.00
ord. For each glyph, the target is the mapped tone of the area
ne line and the next, the candidate
e card, which only covers a tenth of the sheet, |
sh more than 0.6 font sizes are bridged, segmen
or the full text layout-reconstructing ex
moby-dick.reconf.mjs`). Three findin
text is in the file; the proof extracts
s, sizes and light colours, Tesseract
haracters can coincide, exact text equal
4 %) | | Poppler `pdftotext` (default 7
| Poppler `pdftotext -r 600` | 10 of 10
from `pdftotext -bbox-layout`, and MuPDF fo
rs in an ordinary font shows the sa
out 99 % of the words. - **PDF extra
ers and whole pages instead of crops. - Other OCR
art can be seen A Typographics Art PDF carries the whole te
back exactly in every case. ## Where the bytes are Chrom
range of weights, width, size or colour starts
racter, *runs* the style changes (`pdfStats` and `estimate
000 of the type size; at 600 dpi no pixel changed by more th
stitution –9 %, Basic Law –80 %) | lossless (object str
l rows include the slimming steps above; from "one width"
3 MB | 193 | 0.939 / 0.923 | 0.25 % | | without accessibility ta
945 | 3.32 %) | 2 sizes, one width | 1.92 MB | 40 | 0.949 / 0.935
8 | 4 | 0.951 / 0.951 | 3.40 % | | 1 weight, 1 size, one width: onl
ur step 24) | 0.73 MB | 22 | 0.995 / 0.995 | 0.80 % | | without acc
962 | 2.82 | 2.20 % | | 2 sizes, one width | 0.55 MB | 12 | 0.995
0.979 | 2.72 % | | 1 weight, 1 size, one
carries the brightness. ** Two weigh
f 24 save 4 %; step 48 looks the same.
the texture of mixed sizes is a choic
essage that yields the holder of a key
(soft hyphens, direction marks) rem
t of chosen positions. 3. **Fingerpr
n the positions as variable-length inte
g-secret.mjs hide work.pdf --secret "M
ey-stdin`) or from the environment variable `TG_S
cts, and what it does not. **The work never not
ey every time, and repeated letters of the secret p
hat does not occur in the public text cannot be hid
One PDF guilt: after a character drawn from a fall
nd): the work contains its declared text and noth
rt protects data, without reading the cryptograp
esponents behind it are (encrypted-data.md
-and-keys.de.md). ## 1. In one minute - **Ti
public or permanent, the real protection is a
site loads nothing from third parties, and
ryptographer has reviewed it. If losing o
very words, or hex) | The one backup m
```